



Society of Petroleum Engineers

**SPE-229093-MS**

## **Microscale Coupled Simulation of Fracture Propagation and Mineral Weak Interfaces in Enhanced Geothermal Reservoirs: A Case Study of the Gonghe Basin, China**

J. Li, School of Ocean and Earth Science, Tongji University, China / State Key Laboratory of Marine Geology, Tongji University, China; L. Deng, CNPC Chuanqing Drilling Engineering Company Ltd, China; Y. Gou, China National Logging Corporation, China; Z. Zhao, North China Institute of Science and Technology, China; X. Jia, Engineering Technology Research Institute of PetroChina Southwest Oil and Gas Field Company, China; Y. Meng, School of Ocean and Earth Science, Tongji University, China / State Key Laboratory of Marine Geology, Tongji University, China; Y. Li, Liaoning University, China; J. Tang, School of Ocean and Earth Science, Tongji University, China / State Key Laboratory of Marine Geology, Tongji University, China

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229093-MS](https://doi.org/10.2118/229093-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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### **Abstract**

This study focuses on the microscale mechanisms of fracture network development in enhanced geothermal systems (EGS), emphasizing the role of mineralogical heterogeneity and weak interfacial zones. Using the dry hot rock (DHR) reservoir in the Gonghe Basin, Qinghai Province, China, as a case study, we aim to bridge the gap between core-scale observations and microscale mechanical behaviors to support efficient and sustainable geothermal energy extraction.

Our workflow integrates experimental observations and numerical simulations. First, granite cores from the Gonghe Basin were analyzed using thin-section imaging and scanning electron microscopy to quantify mineral composition and weak plane distribution. Mechanical properties at the grain scale were calibrated through instrumented nano-indentation. These data were used to construct a finite element model with embedded Cohesive Zone Method (CZM) elements along mineral boundaries. The model captures debonding and fracture coalescence across clay zones, quartz interfaces, and pre-existing microcracks. Simulations were conducted under geothermal stress regimes to analyze the effects of Young's modulus heterogeneity, interfacial toughness, and pressure gradients.

The results demonstrate that fracture initiation and propagation in EGS reservoirs are highly dependent on microscale mineral arrangement and interfacial mechanical contrast. Weak zones, especially those formed by feldspar-altered contacts or clay inclusions, serve as preferred crack paths. Fractures exhibit deflection, branching, and coalescence patterns that align with mineral texture anisotropy, leading to enhanced complexity and connectivity of the network. Thermal loading promotes intergranular fracture opening and accelerates weak-interface softening. Our damage and stress field simulations show strong correlation with experimental fracture surfaces and acoustic emission zones. Overall, this multi-scale modeling framework

provides a reliable means to predict fracture geometry evolution and optimize stimulation protocols in complex geothermal reservoirs.

By integrating nano-indentation data, mineralogical maps, and cohesive zone mechanics into a unified microscale framework, this study offers a novel approach to modeling fracture propagation in dry hot rock geothermal systems. The method improves the fidelity of fracture prediction and helps guide tailored stimulation designs for sustainable geothermal exploitation in mineral-heterogeneous environments.

## Introduction

Geothermal energy is increasingly recognized as a reliable low-carbon energy source capable of providing stable baseload power, unlike intermittent renewables such as wind and solar. Enhanced geothermal systems (EGS) that target hot dry rock (HDR) in crystalline basement formations are particularly attractive because hydraulic stimulation can generate interconnected fracture networks and unlock large subsurface heat reservoirs, thereby contributing to long-term energy security and decarbonization goals. International assessments emphasize this strategic potential: the International Renewable Energy Agency highlights geothermal as indispensable for future power stability (IRENA, 2023), the U.S. Department of Energy projects that technological advances could enable widespread EGS deployment by 2050 (DOE, 2019), and the Massachusetts Institute of Technology estimated that HDR resources could provide hundreds of gigawatts of baseload electricity if effectively developed (Tester et al., 2006). In China, the Gonghe Basin on the northeastern Tibetan Plateau has become the key HDR demonstration area, with high heat-flow anomalies, granitic intrusions, and combined radiogenic and mantle heating creating favorable geological conditions (Zhang et al., 2021; Yang et al., 2023; Liu et al., 2023). Field injection–production experiments there have confirmed the feasibility of long-term circulation but also revealed strong sensitivity to reservoir heterogeneity, natural fractures, and stress redistribution (Zhu et al., 2024; Zhang et al., 2022). The Gonghe granite, dominated by quartz, feldspar, and mica, is enriched in radiogenic elements that further elevate its geothermal potential (Yang et al., 2023).

A central challenge for EGS is understanding hydraulic fracture propagation in crystalline granite, where fracture geometry and connectivity are controlled by mineral heterogeneity, weak grain boundaries, and interactions with natural fractures. Numerical modeling has become essential to investigate these mechanisms. Continuum approaches such as the Finite Element Method (FEM) and Extended FEM allow simulation of fracture growth and branching (Belytschko et al., 1999), while the phase-field method has gained attention for representing complex fracture topologies (Miehe et al., 2010). Boundary element and displacement discontinuity methods provide efficient large-scale fracture network analysis (Crouch and Starfield, 1983), and lattice or peridynamics models capture crack nucleation and dynamic propagation (Ostoja-Starzewski, 2002; Silling, 2000). However, these continuum methods often assume homogeneity and fail to capture mineral-scale processes.

The Discrete Element Method (DEM) offers a powerful alternative for crystalline rocks, where fractures emerge naturally from particle bond breakage. First introduced by Cundall and Strack (1979) and extended with the bonded-particle model (Potyondy and Cundall, 2004), DEM implemented in PFC2D/3D (Itasca, 2025) has successfully reproduced hydraulic fracturing experiments, including breakdown pressures and acoustic emission patterns (Al-Busaidi et al., 2005; Han et al., 2018; Jiang et al., 2023). DEM further allows explicit representation of mineral heterogeneity, with quartz, feldspar, and mica modeled with distinct mechanical properties and grain boundaries represented as weak planes. Recent studies show that such heterogeneity induces fracture deflection, branching, and network complexity (Mo et al., 2023; Zhang et al., 2023; Gao et al., 2023; Liu et al., 2023a; Pan et al., 2023), which is particularly relevant to Gonghe granite. Another critical aspect is the interaction between hydraulic and natural fractures. Laboratory and theoretical studies established criteria for crossing or diversion (Renshaw and Pollard, 1995; Gu and Weng, 2011), later refined to include cohesive–frictional properties and fluid pressure (Kolawole et al., 2020; Abe

et al., 2019). DEM complements these approaches by directly simulating fracture intersections, showing that stress difference, angle, interface strength, and injection rate jointly determine outcomes (Han et al., 2018; Liu et al., 2023a). In fractured HDR systems, such mechanics control whether stimulation produces isolated fractures or interconnected networks.

Despite advances, significant challenges remain. Continuum models often oversimplify granite as homogeneous, while DEM calibration between laboratory and particle-scale parameters remains non-trivial (Mo et al., 2023; Zhang et al., 2023). Most interaction criteria are derived under simplified conditions and lack validation in geothermal settings. Finally, bridging scales is unresolved: grain-scale DEM insights are rarely integrated into reservoir simulators, while large-scale models neglect mineral-scale processes essential for fracture initiation.

The present study addresses these gaps through two hydro-mechanically coupled PFC2D simulations of Gonghe granite. The first explores fracture propagation in a grain-based synthetic granite incorporating mineral heterogeneity, demonstrating the influence of weak interfaces and mineral contrasts. The second introduces a pre-existing natural fracture represented as a non-orthogonal cohesive–frictional discontinuity to analyze fracture–fracture interactions. Although exploratory, these simulations emphasize the underexplored role of mineral weak planes and natural fractures in Chinese HDR reservoirs. The novelty lies in explicitly linking microscale heterogeneity with fracture propagation using DEM, thereby complementing homogeneous assumptions in prior studies and offering mechanistic insights to inform reservoir-scale stimulation design

## Geological Background

The Gonghe Basin in Qinghai Province, located on the northeastern margin of the Tibetan Plateau, is one of the most important sites for hot dry rock (HDR) exploration and demonstration in China. Situated between the Qaidam and West Qinling orogenic belts, the basin has experienced crustal thickening, mantle upwelling, and radiogenic heat enrichment, producing high geothermal gradients favorable for HDR development (Zhang et al., 2021; Yang et al., 2023). Geological surveys indicate that Triassic granites dominate the basement, supplemented by basaltic and metamorphic units that provide both thermal reservoirs and structural framework (Xu et al., 2019).

The granitic basement, mainly composed of quartz, feldspar, and mica, is enriched in U, Th, and K, which enhances radiogenic heating. Locally, these granitoids are overlain by basaltic flows and Quaternary sediments that serve as effective caprocks, preserving geothermal heat (Liu et al., 2023). Exploration drilling has shown temperatures exceeding 200 °C at depths of ~3700 m, confirming the basin's high enthalpy potential (Zhu et al., 2024).

In 2017, the China Geological Survey (CGS) drilled the first HDR borehole in Gonghe granite to 3705 m, recording bottom-hole temperatures of 236 °C. This led to the HDR Exploration and Production Demonstration Project, launched in 2019. With advances in directional drilling, hydraulic stimulation, and microseismic monitoring, the first power generation test was successfully completed in 2021, marking a milestone in China's HDR utilization (Zhang et al., 2022) (Fig. 1a). More recently, three-dimensional geological modeling has delineated the extent of HDR resources, showing large continuous granite bodies at depths >3000 m, providing key data for resource estimation and site selection (Zhu et al., 2023) (Fig. 1b).

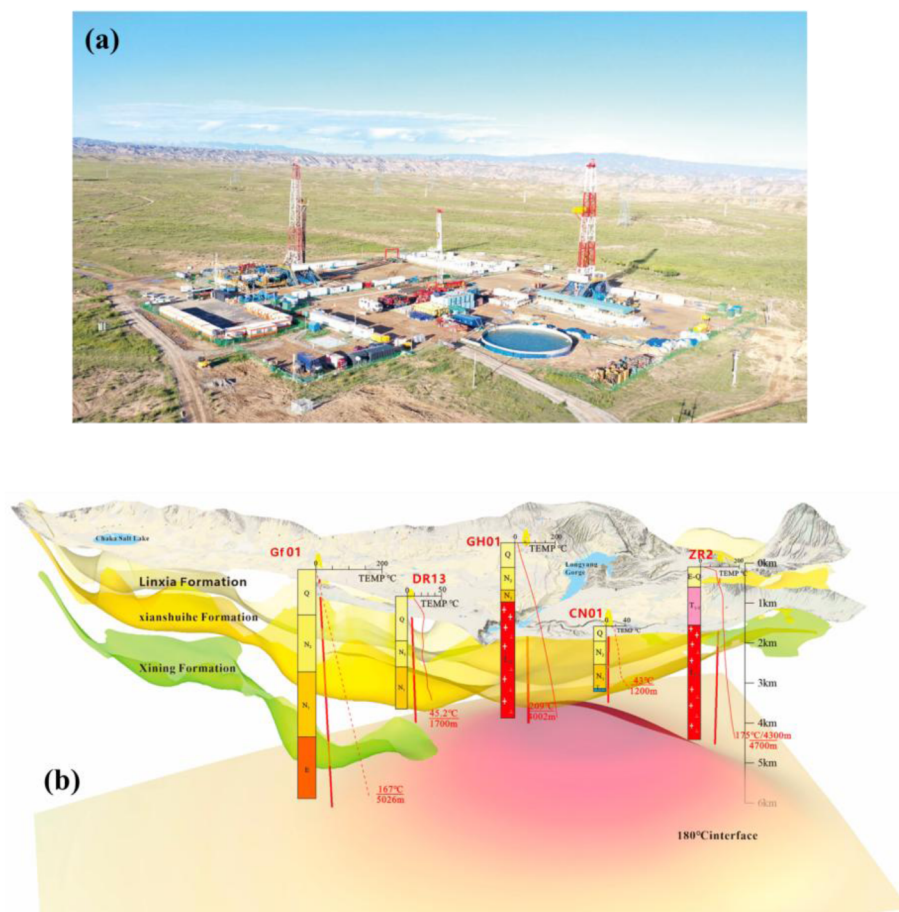


Figure 1—Hot dry rock (HDR) resources in the Gonghe Basin: (a) Site view of the HDR Exploration and Production Demonstration Project, Qinghai Province, China (from Zhang et al., 2022); (b) Three-dimensional geological model of the distribution of the HDR body in the Gonghe Basin (from Zhu et al., 2023).

Overall, the unique tectonic setting, radiogenic granites, and localized basaltic cover make the Gonghe Basin one of China's most promising HDR prospects and a natural laboratory for investigating hydraulic fracturing and heat extraction in crystalline rocks.

## Methodology

### Numerical Modeling Framework

In this study, the hydro-mechanical coupling process of hydraulic fracture propagation in the granitic reservoir of the Gonghe Basin was simulated using the Particle Flow Code in two dimensions (PFC2D), which is based on the discrete element method (DEM). In PFC2D, the rock matrix is represented as an assembly of circular particles bonded through contact models, and deformation or fracturing is realized by progressive breakage of these bonds under applied stress and fluid pressure. This approach is particularly suitable for crystalline rocks such as granite, as it allows for explicit representation of mineral-scale heterogeneity and weak interfaces that strongly influence fracture initiation and growth.

The contact mechanics in PFC2D include both normal and shear stiffness, as well as tensile and shear strengths, to describe elastic deformation and failure. To account for mineralogical heterogeneity in Gonghe granite, particles are assigned mineral types—primarily quartz, feldspar, and mica—according to petrographic analysis and published compositional data. Different mineral types are given distinct mechanical parameters such as stiffness, strength, and friction angle, reflecting their contrasting physical properties observed in laboratory tests.



Mineral weak interfaces are explicitly incorporated by reducing bond strength and stiffness at contacts between dissimilar minerals. For example, quartz–feldspar and feldspar–mica contacts are modeled with weaker cohesion than quartz–quartz contacts, representing the tendency for fracture deflection or branching along grain boundaries. This enables the model to reproduce preferential crack propagation through weak planes and the influence of mineral distribution on overall fracture geometry.

The model domain is loaded by applying in-situ stress conditions representative of the Gonghe Basin. Fluid injection is imposed at a designated point, generating a pressure gradient that drives fracture initiation and extension. The DEM framework allows new fractures, branching, and coalescence to emerge spontaneously without predefined paths, enabling direct investigation of how granite heterogeneity and natural fracture interfaces control fracture propagation and network development under hydraulic stimulation.

### Governing Equations

The mechanical behavior of each particle in PFC2D follows the laws of motion. For particle  $i$ , Newton's second law governs translational motion, and the rotational equation governs angular motion:

$$m_i \frac{d\mathbf{v}_i}{dt} = \sum_j \mathbf{F}_{ij} + m_i \mathbf{g} \quad (1)$$

$$I_i \frac{d\boldsymbol{\omega}_i}{dt} = \sum_j \mathbf{M}_{ij} \quad (2)$$

where  $m_i$  is the particle mass,  $I_i$  is the moment of inertia,  $\mathbf{v}_i$  is the translational velocity,  $\boldsymbol{\omega}_i$  is the angular velocity,  $\mathbf{F}_{ij}$  is the contact force exerted by neighboring particle  $j$ ,  $\mathbf{M}_{ij}$  is the corresponding contact moment, and  $\mathbf{g}$  is gravitational acceleration.

The contact force is decomposed into normal and shear components:

$$\mathbf{F}_{ij} = \mathbf{F}_{n,ij} + \mathbf{F}_{s,ij} \quad (3)$$

$$\mathbf{F}_{n,ij} = k_n \delta_n \mathbf{n} \quad (4)$$

$$\mathbf{F}_{s,ij} = k_s \delta_s \mathbf{s} \quad (5)$$

where  $k_n$  and  $k_s$  are the normal and shear stiffnesses,  $\delta_n$  and  $\delta_s$  are the normal and shear displacements, and  $\mathbf{n}$  and  $\mathbf{s}$  are the unit normal and shear vectors at the contact.

Bonded contacts can fail either in tension or shear. Tensile failure occurs when the maximum normal stress exceeds the tensile strength:

$$\sigma_n \geq \sigma_{n,\max} \quad (6)$$

Shear failure occurs when the maximum shear stress exceeds the shear strength, following a Mohr–Coulomb criterion:

$$\tau_s \geq c + \sigma_n \tan \phi \quad (7)$$

where  $c$  is cohesion and  $\phi$  is the friction angle. Once a bond fails, the contact transitions to a frictional sliding contact, allowing the formation of discrete fractures.

### Fluid–Solid Coupling

Fluid flow in the model is simulated using a network of flow channels formed by the contacts between particles. The continuity equation governs the change in fluid volume within each pore:

$$\frac{dV_f}{dt} = \sum_j Q_{ij} + Q_s \quad (8)$$

where  $V_f$  is the fluid volume,  $Q_{ij}$  is the flow rate between pores  $i$  and  $j$ , and  $Q_s$  is the source/sink term representing injection or extraction. The flow rate is calculated using the cubic law for laminar flow between parallel plates:

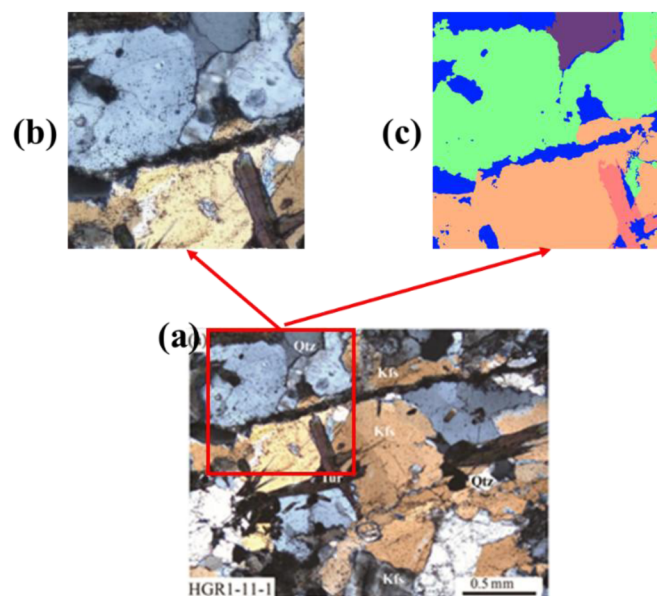
$$Q_{ij} = \frac{b_{ij}^3}{12\mu} \cdot \frac{\Delta p_{ij}}{l_{ij}} \cdot w_{ij} \quad (9)$$

where  $b_{ij}$  is the hydraulic aperture,  $\mu$  is fluid viscosity,  $\Delta p_{ij}$  is the pressure difference,  $l_{ij}$  is the flow path length, and  $w_{ij}$  is the model thickness (out-of-plane width).

Fracture aperture evolution is directly linked to particle displacement: as bonds break and particles separate, the aperture  $b_{ij}$  increases, enhancing local conductivity and altering fluid flow paths. The coupled calculation proceeds by alternately updating particle motion from contact and fluid forces, and fluid pressures from updated apertures, thus achieving two-way coupling between mechanical deformation and fluid flow.

### Microstructural Characterization and Voronoi-Based Geometric Modeling

The microscale mineralogical features of the Gonghe granite were characterized using polarized thin-section microscopy (Fig. 3a). High-resolution micrographs were obtained to capture the distribution, morphology, and intergranular contacts of mineral phases. A local region of interest was magnified to highlight characteristic mineral textures, cleavage planes, and weak boundaries (Fig. 3b). Image segmentation was then conducted in Avizo software, producing a color-coded classification map of distinct mineral types (Fig. 3c). The granite sample is dominated by quartz, K-feldspar, and plagioclase, with minor accessory minerals such as biotite and tourmaline, consistent with previous petrographic studies of HDR reservoirs in the Gonghe Basin. The grains are generally angular, showing clear boundaries and weak interfaces, which play an important role in crack nucleation and propagation.



**Figure 3—Microscale mineral classification and segmentation of Gonghe granite thin section (modified after Yun et al., 2020): (a) Representative polarized light thin-section image showing mineral assemblages and grain boundaries; (b) Magnified sub-region highlighting mineral textures and weak contacts; (c) Color-segmented classification map of mineral phases, used as the basis for Voronoi tessellation and discrete element modeling.**

To construct the numerical model, the segmented mineral map was used as the primary input for Voronoi-based geometric modeling. Each identified mineral grain centroid was treated as a seed point, and a Voronoi

tessellation was generated to define the polygonal boundaries of individual grains. The Voronoi region  $v_i$  associated with the  $i$ -th seed point  $p_i$  is defined as:

$$v_i = \left\{ v \mid \|v - p_i\|_2 \leq \|v - p_j\|_2, \forall j \neq i \right\} \quad (10)$$

where  $v$  is any point in the plane,  $p_i$  is the  $i$ -th Voronoi seed, and  $\|\cdot\|_2$  denotes the Euclidean distance.

The polygonal grain boundaries were mapped to the PFC2D model domain, and rblock scripts were generated to assign mineral types and their associated mechanical parameters. Quartz grains were given relatively high stiffness and strength, feldspars intermediate properties, and mica weaker stiffness and cohesion. Interfaces between dissimilar minerals were modeled with reduced bond strengths to represent weak planes along which cracks preferentially propagate. This Voronoi-based geometric modeling ensures that the discrete element simulations realistically incorporate mineralogical heterogeneity and weak interfaces, thereby enabling mechanistic investigation of fracture propagation in crystalline HDR reservoirs of the Gonghe Basin.

## Model Construction and Boundary Conditions

**Table 1—Model parameters used in the PFC2D hydraulic fracturing simulations (this study)**

| Parameter                                        | Value                | Unit              |
|--------------------------------------------------|----------------------|-------------------|
| Model dimensions                                 | $0.2 \times 0.2$     | m                 |
| Porosity                                         | 0.08                 | —                 |
| Particle density                                 | 2650                 | kg/m <sup>3</sup> |
| Contact normal stiffness                         | $1.0 \times 10^9$    | Pa                |
| Shear-to-normal stiffness ratio (contacts)       | 2.5                  | —                 |
| Contact friction coefficient                     | 0.5                  | —                 |
| Parallel bond normal stiffness                   | $8.0 \times 10^9$    | Pa                |
| Shear-to-normal stiffness ratio (parallel bonds) | 1.8                  | —                 |
| Bond tensile strength                            | 20                   | MPa               |
| Bond shear strength                              | 50                   | MPa               |
| Equivalent permeability                          | $1.0 \times 10^{-3}$ | m <sup>2</sup>    |
| Fluid bulk modulus                               | $2.0 \times 10^9$    | Pa                |
| Initial pore pressure                            | 50                   | MPa               |
| Fluid–solid calculation time step                | $5 \times 10^{-5}$   | s                 |
| Inlet flow rate                                  | $1.0 \times 10^{-5}$ | m <sup>3</sup> /s |
| Target stress in X direction                     | -10                  | MPa               |
| Target stress in Y direction                     | -5                   | MPa               |

A two-dimensional particle flow model was constructed in PFC6.0 to simulate the hydraulic fracturing process incorporating mineral weak interfaces. The computational domain was defined as  $0.2 \text{ m} \times 0.2 \text{ m}$ . The spatial distribution of different minerals was identified from polarized light microscope images, and corresponding rigid blocks were generated to match the segmented mineral boundaries. Within each block, particles were randomly generated within a specified radius range to achieve a target porosity of 0.08, with particle density set to 2650 kg/m<sup>3</sup>.

The particle contact behavior followed a linear contact model, with a normal stiffness of  $1.0 \times 10^9$  Pa, a shear-to-normal stiffness ratio of 2.5, and a friction coefficient of 0.5. Bonding between particles

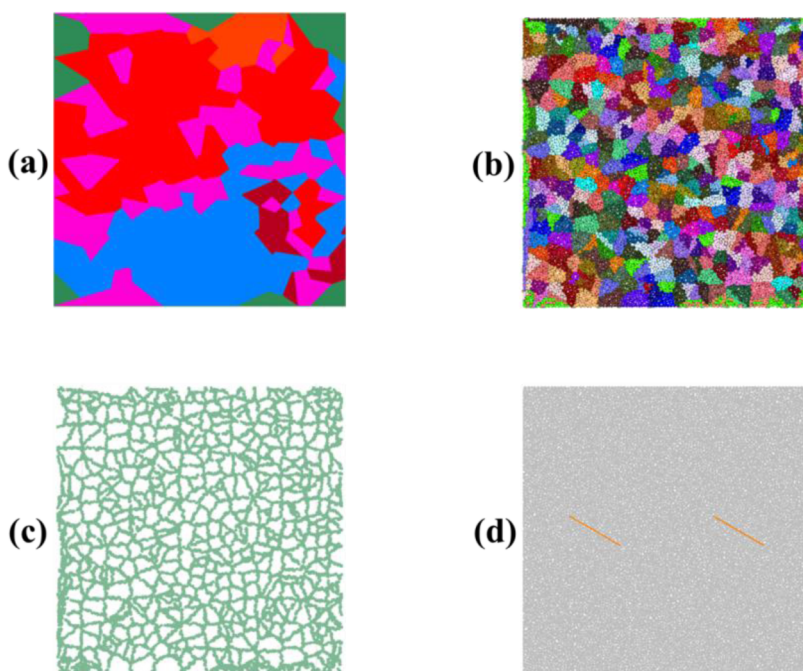
was modeled using the parallel bond model to simulate cementation, with a normal stiffness of  $8.0 \times 10^9$  Pa, a shear-to-normal stiffness ratio of 1.8, and tensile and shear bond strengths of 20 MPa and 50 MPa, respectively. Interface groups representing mineral weak boundaries were assigned lower strength parameters to reflect reduced cohesion along mineral contacts.

The seepage field was simulated using fully coupled fluid–solid interaction calculations. The equivalent permeability was set to  $1.0 \times 10^{-3}$  m<sup>2</sup>, the fluid bulk modulus to  $2.0 \times 10^9$  Pa, and the initial pore pressure to 50 MPa. The fluid–solid calculation used a time step of  $5 \times 10^{-5}$  s. A constant inlet flow rate of  $1.0 \times 10^{-5}$  m<sup>3</sup>/s was applied, and the hydraulic aperture of flow channels was dynamically updated during fracture propagation.

All four boundaries were modeled as rigid walls, with servo-controlled loading applied to maintain target stresses of  $-10$  MPa in the X-direction and  $-5$  MPa in the Y-direction. Stress control was achieved by real-time adjustment of wall velocities to maintain the target boundary stresses. Once the target stresses were reached and mechanical equilibrium was established, the simulation proceeded to the fracture propagation and seepage evolution stage.

### Discrete Element Assembly and Bonding Structure

The geometric model derived from Voronoi tessellation of mineral distributions was imported into PFC2D as rigid blocks using the RBlock command (Fig. 4a). Each block corresponds to a single mineral grain identified in the image segmentation stage, with block group IDs assigned according to mineral type. This ensured that the microstructural heterogeneity of the Gonghe granite was faithfully represented in the numerical domain.



**Figure 4—Construction of the discrete element model for hydraulic fracturing simulation: (a) Rigid block geometry imported into PFC2D based on Voronoi tessellation of mineral distribution; (b) Discrete particle assembly generated within each rigid block to match the target porosity; (c) Bond network established between particles, with reduced strength at mineral interfaces to represent weak boundaries; (d) Predefined weak-plane model with inclined discontinuities introduced for fracture–fracture interaction analysis (this study).**

Within each rigid block, a discrete particle assembly was generated (Fig. 4b). Particle radii were assigned within a predefined range to achieve the target porosity, and particle packing was conducted in a manner that preserved the geometry of the rigid block boundaries. This procedure allowed each mineral grain to



be internally represented by bonded particles, enabling the simulation of both intra-grain and inter-grain fracture development.

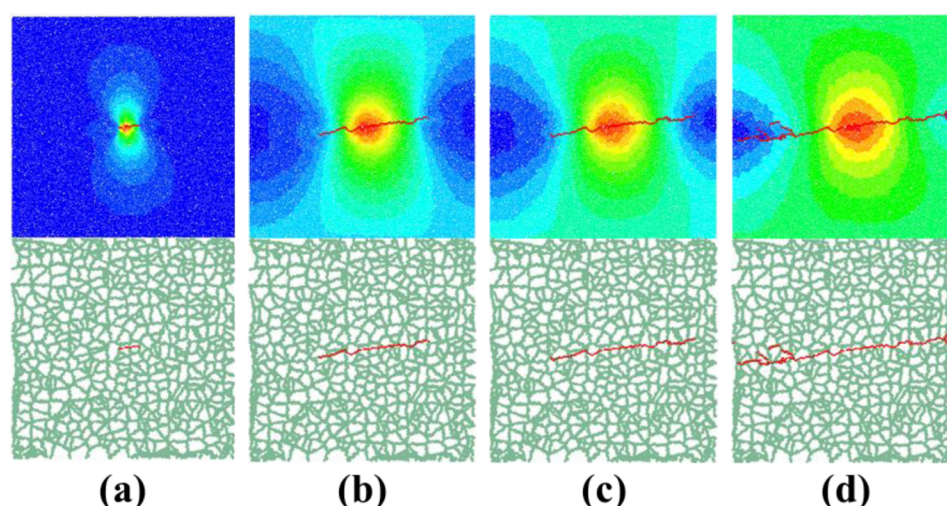
Once the particle assembly was generated, bonding between particles was established using the parallel bond model (Fig. 4c). Bonds within the same mineral block were assigned stiffness and strength parameters consistent with the mineral's mechanical properties, whereas bonds across mineral interfaces were assigned reduced strength to simulate weak mineral boundaries, allowing fractures to preferentially initiate and propagate along these contacts under hydraulic and mechanical loading.

On this basis, a weak-plane model was introduced to analyze the interaction between induced fractures and pre-existing structures (Fig. 4d). Two parallel discontinuities inclined at  $30^\circ$  were embedded in the model with a fixed aperture of 0.2 mm. Their mechanical behavior was controlled by the smooth-joint contact model, in which the normal and shear stiffnesses were set to 500 GPa and 60 GPa, respectively. The joints were assigned a friction coefficient of 0.70, tensile and shear bond strengths of 2.0, and a residual friction angle of  $40^\circ$ . The discontinuities were installed along the centerline, extending 0.1 mm on each side to ensure effective recognition and constraint of contact pairs. After initialization cycles and numerical balancing, the model was saved for subsequent loading and hydro-mechanical response simulations.

## Results and Discussions

### Fracture Propagation Evolution

The evolution of fracture initiation and propagation in the Gonghe granite model under coupled hydro-mechanical loading is shown in Fig. 5, which illustrates both stress field distribution (upper panels) and fracture geometry (lower panels) at four key simulation stages.



**Figure 5—Evolution of fracture propagation and stress field in the Gonghe granite model under coupled hydro-mechanical loading: (a) cycle 2,000, initial fracture initiation and symmetric stress concentration; (b) cycle 6,000, stable fracture extension along the maximum principal stress direction; (c) cycle 10,000, emergence of secondary cracks and stress redistribution; (d) cycle 14,000, fracture branching and deflection influenced by mineral weak interfaces (this study).**

At the early stage (cycle 2,000; Fig. 5a), fluid injection produces a localized high-pressure zone near the injection point, with stress concentration developing at nearby bonded contacts. A short tensile fracture initiates preferentially along a quartz–feldspar boundary where tensile strength is lowest. The stress field exhibits a nearly symmetric pattern, consistent with early Mode I fracture opening.

By cycle 6,000 (Fig. 5b), the main fracture extends steadily in the direction of maximum principal stress. Redistribution of the surrounding stress field creates tensile stress zones ahead of the crack tip.

The propagation path remains relatively straight, reflecting the dominance of the stiff quartz framework in guiding fracture growth.

At cycle 10,000 (Fig. 5c), the fracture length increases significantly, and secondary microcracks appear at oblique angles, triggered by heterogeneity in mineral contacts and local stiffness contrasts. The stress field becomes more complex, with lobes extending laterally, suggesting interaction between the primary fracture and weaker mica- or clay-bearing interfaces.

By cycle 14,000 (Fig. 5d), the fracture network shows pronounced branching and deflection, with segments deviating toward zones of reduced strength. The stress distribution near the crack tip becomes asymmetric, reflecting anisotropic mineralogical control. These results highlight the crucial role of mineral weak interfaces and local stress perturbations in governing fracture geometry, branching, and connectivity in crystalline HDR reservoirs.

### Evolution of Fracture, Pressure, and Stress

The evolution of fracture count with simulation step is shown in Fig. 6. During the initial stage (Step  $\approx$  10,000–14,000), only a few fractures nucleate around the injection point, reflecting localized damage initiation along mineral contacts. Most injected fluid is accommodated by pore expansion and early fracture opening, consistent with the nucleation phase. Between Step  $\approx$  14,000–20,000, fracture count increases nearly linearly, indicating stable propagation of the dominant fractures along the maximum principal stress direction. Limited branching occurs, and the system remains hydraulically stable. At the later stage (Step  $>$  20,000), the fracture number rises rapidly, marking the onset of unstable propagation. This acceleration reflects enhanced interaction among cracks, as hydraulic connectivity develops along weak mineral boundaries and secondary fractures coalesce into a connected network.

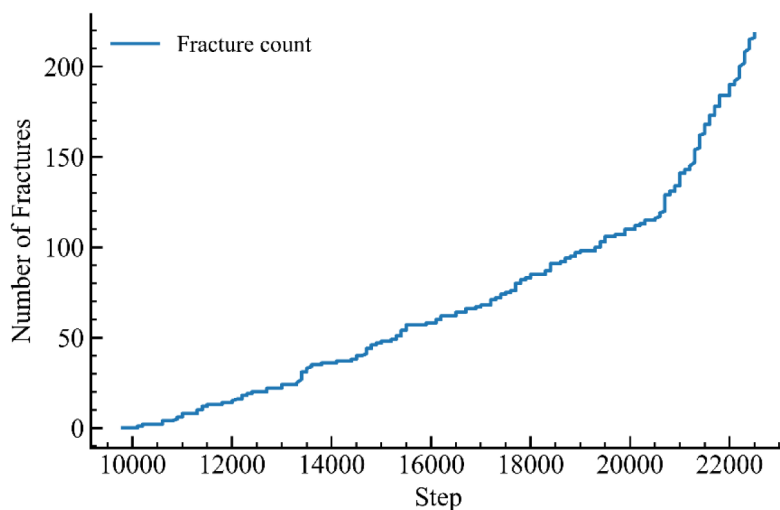
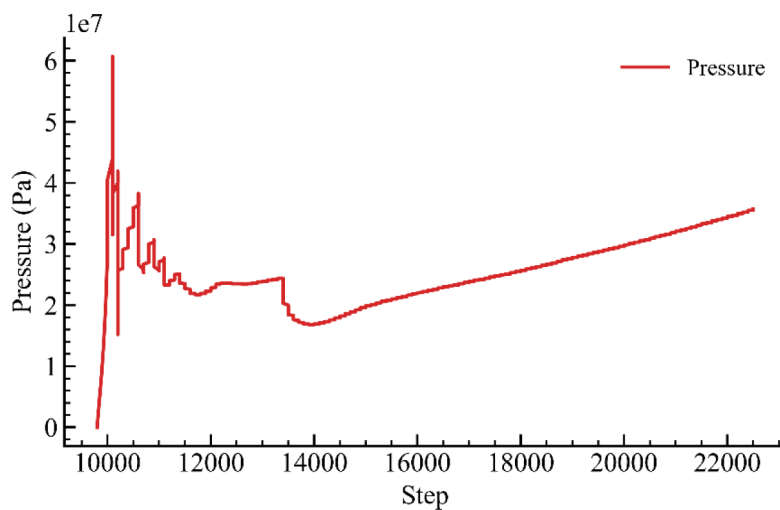


Figure 6—Evolution of fracture number with simulation step: three distinct stages are observed—slow nucleation and growth, steady linear propagation, and accelerated branching in granite with mineral weak interfaces (this study).

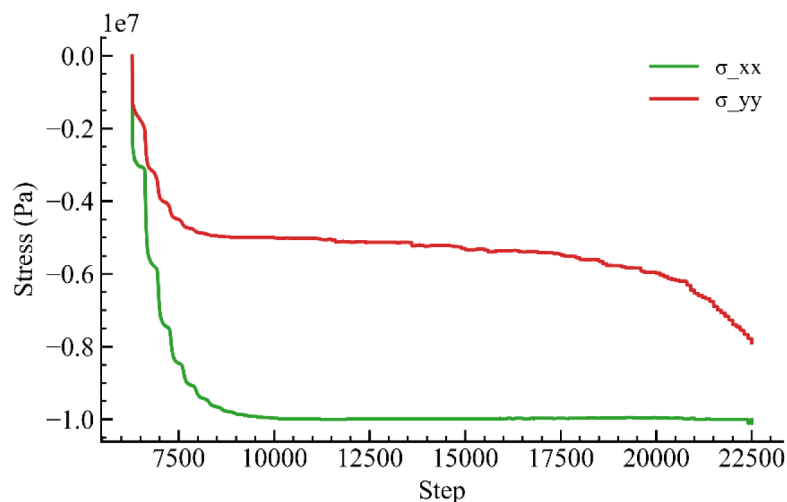
Fig. 7 presents the evolution of injection pressure during the simulation. Initially, the pressure increases sharply as the system is loaded and fluid is injected into an intact rock matrix. The first pressure peak corresponds to the breakdown pressure, at which the initial fractures form. Following breakdown, the pressure shows a series of fluctuations due to the intermittent extension of fractures and the creation of new flow pathways. These fluctuations are particularly evident between Step  $\approx$  10,000–14,000, coinciding with the early fracture growth stage seen in Fig. 6. After Step  $\approx$  15,000, pressure stabilizes at a lower value as newly formed fractures enhance permeability, allowing the injected fluid to dissipate more efficiently. However, toward the end of the simulation (Step  $>$  20,000), the pressure begins to rise gradually again,

likely due to increased flow resistance caused by partial closure or tortuosity of fracture pathways as they propagate into more heterogeneous mineral zones.



**Figure 7—Variation of injection pressure with simulation step: a rapid rise to breakdown pressure is followed by oscillations during fracture extension, stabilization during steady growth, and gradual increase during late-stage branching in heterogeneous granite (this study).**

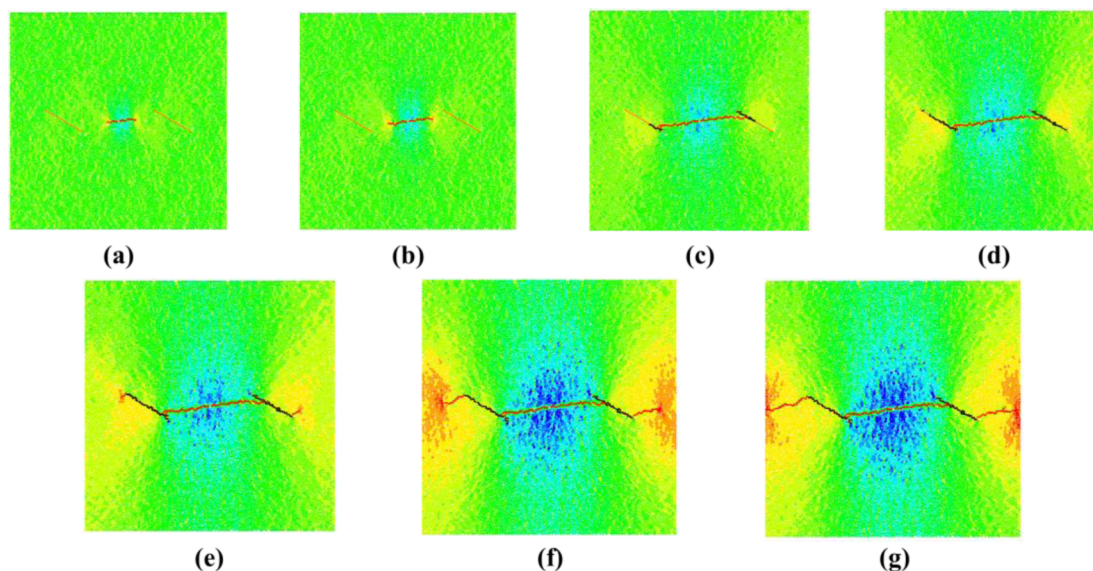
Fig. 8 illustrates the temporal evolution of the principal stresses  $\sigma_{xx}$  and  $\sigma_{yy}$  during hydraulic fracturing. At the onset of fluid injection, both stress components decrease sharply due to stress redistribution around the injection point and crack tip. The reduction in  $\sigma_{xx}$  is more pronounced than in  $\sigma_{yy}$ , indicating stronger stress relaxation along the maximum principal stress direction that governs fracture propagation. After Step  $\approx 10,000$ , both stresses enter a quasi-steady state, corresponding to the stable propagation phase observed in Figs. 6 and 7. During this stage,  $\sigma_{xx}$  remains nearly constant while  $\sigma_{yy}$  shows a gradual decline, reflecting minor adjustments in the stress field. Toward the late stage (Step  $> 20,000$ ),  $\sigma_{yy}$  exhibits a further drop, coinciding with accelerated fracture branching and coalescence. This indicates that fracture network expansion induces additional stress relief perpendicular to the main fracture direction, consistent with the transition to a more complex fracture system in the heterogeneous granite.



**Figure 8—Temporal evolution of principal stresses  $\sigma_{xx}$  and  $\sigma_{yy}$ : rapid stress drop after injection, quasi-steady behavior during stable fracture growth, and further decline in  $\sigma_{yy}$  during accelerated network expansion (this study).**

## Fracture Propagation with Weak Interfaces

The fracture propagation process in the Gonghe granite model incorporating weak interfaces is shown in Fig. 9. At the early stage (cycle 1,000; Fig. 9a), fluid injection generates localized tensile stress concentration near the borehole, initiating fractures preferentially along pre-existing weak interfaces. By cycle 3,000 (Fig. 9b), the fracture segments extend stably, with stress redistribution forming symmetric tensile zones on both sides of the injection point. At cycle 5,000 (Fig. 9c), fractures continue to propagate along the weak boundaries, with minor deflections at mineral contacts. By cycle 7,000 (Fig. 9d), a dominant fracture path emerges, strongly controlled by the orientation of the weak interfaces, while stress relaxation zones form ahead of the advancing tips. At cycle 9,000 (Fig. 9e), branching occurs as secondary cracks nucleate along adjacent weak planes. As propagation continues to cycle 11,000 (Fig. 9f), the fracture network becomes more complex, with secondary cracks merging into the main fracture path, leading to heterogeneous stress redistribution. Finally, at cycle 11,452 (Fig. 9g), the fracture system evolves into a connected network, exhibiting multiple branches, deflections, and asymmetric growth patterns driven by the distribution and orientation of weak interfaces. These results highlight that mineral-scale weak planes exert a primary control on fracture initiation, branching, and network connectivity under hydro-mechanical coupling in crystalline HDR reservoirs.

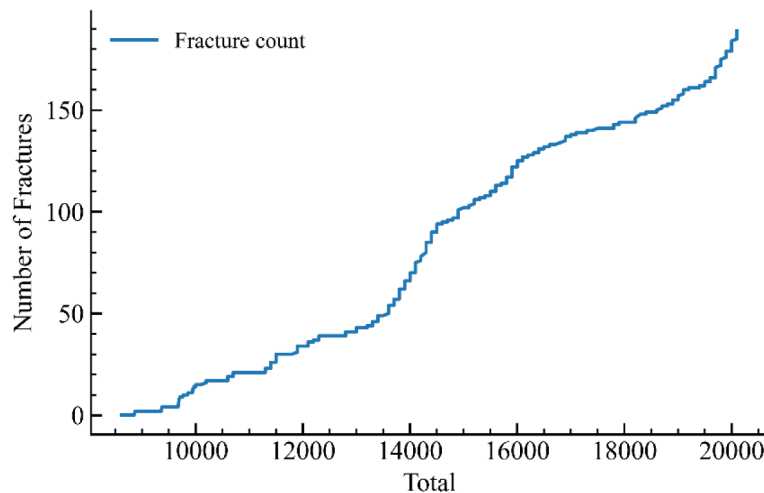


**Figure 9—Evolution of fracture propagation and stress field in the Gonghe granite model with weak interfaces under coupled hydro-mechanical loading: (a) cycle 1,000, fracture initiation along weak interfaces near the injection point; (b) cycle 3,000, stable extension with symmetric stress redistribution; (c) cycle 5,000, propagation with minor deflection at mineral boundaries; (d) cycle 7,000, dominant path guided by weak planes; (e) cycle 9,000, onset of branching along adjacent weak interfaces; (f) cycle 11,000, secondary cracks coalescing with the main fracture; (g) cycle 11,452, formation of a connected branched fracture network (this study).**

## Evolution of Fracture, Pressure, and Stress with Weak Interfaces

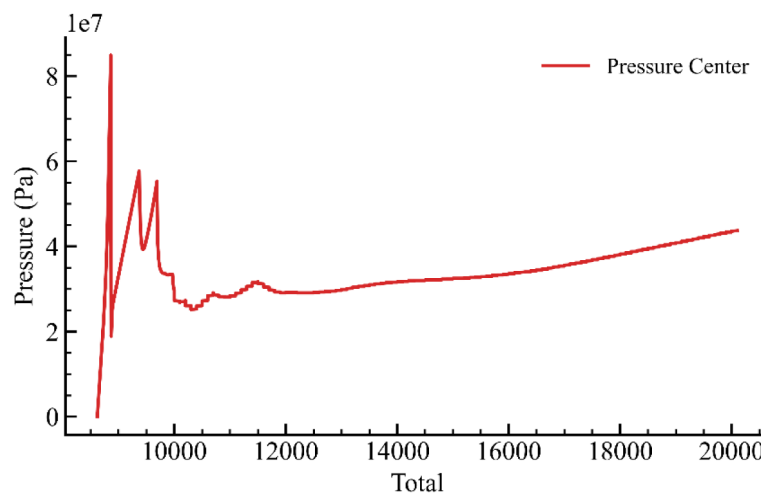
The coupled hydro-mechanical simulations incorporating weak mineral interfaces demonstrate distinct evolution patterns for fracture development, injection pressure, and stress redistribution. The fracture number increases in three characteristic stages (Fig. 10). Initially, only limited cracks nucleate around the injection point, as fracture growth is confined to isolated weak planes. During the intermediate stage, fracture propagation follows the dominant stress direction, guided by mineral interfaces, leading to nearly linear growth in fracture count. In the late stage, interaction between multiple weak planes promotes branching and coalescence, resulting in accelerated network development.





**Figure 10—Evolution of fracture number with simulation step: the fracture count in granite with weak interfaces increases in three stages: slow nucleation, steady extension dominated by interface-guided cracks, and accelerated branching when multiple weak planes interact (this study).**

Injection pressure evolution (Fig. 11) further highlights the controlling role of weak interfaces. A sharp increase to breakdown pressure is followed by strong oscillations during early fracture-interface interactions. Once stable propagation is established, pressure decreases and remains relatively steady, reflecting enhanced fluid conductivity. At later stages, pressure gradually increases again due to tortuous flow paths and increased resistance as the network expands.



**Figure 11—Variation of injection pressure with simulation step: pressure rises sharply to breakdown, then exhibits pronounced oscillations linked to fracture–interface interactions, stabilizes during steady extension, and gradually increases again during late-stage branching along weak planes (this study).**

Principal stress evolution (Fig. 12) reveals that  $\sigma_{xx}$  drops rapidly after injection and stabilizes once the main fracture aligns with the weak planes, whereas  $\sigma_{yy}$  continues to decline gradually. This indicates sustained stress redistribution perpendicular to the fracture orientation, consistent with the accelerated fracture branching stage. Collectively, these results show that weak interfaces not only reduce breakdown pressure but also control fracture geometry, branching intensity, and long-term stress evolution.

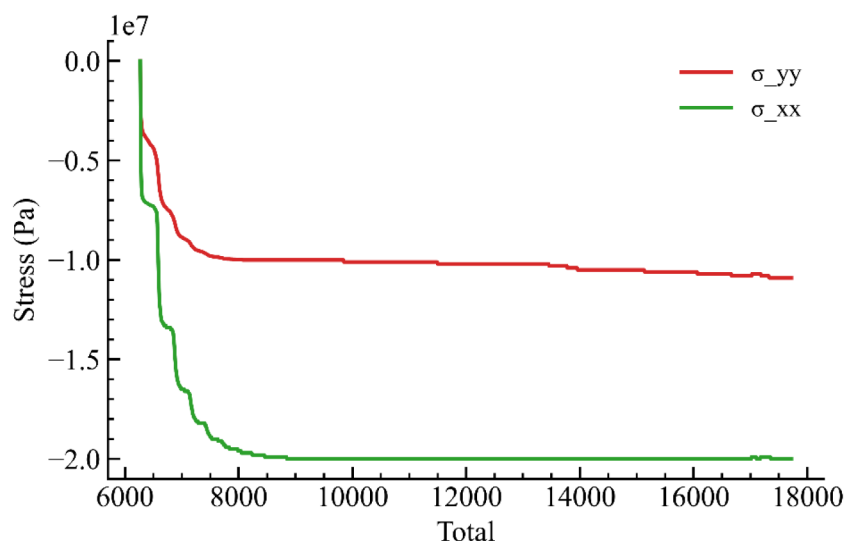


Figure 12—Temporal evolution of principal stresses  $\sigma_{xx}$  and  $\sigma_{yy}$ : both stresses drop rapidly after injection;  $\sigma_{xx}$  stabilizes earlier, while  $\sigma_{yy}$  shows a continuous decline during fracture–interface coalescence and branching, indicating sustained stress redistribution (this study).

## Conclusions and Recommendations

This study developed a hydro-mechanically coupled PFC2D model to investigate fracture propagation in granitic hot dry rock reservoirs of the Gonghe Basin, with explicit consideration of mineral heterogeneity and weak interfaces. By incorporating polarized light microstructural images into a Voronoi-based discrete element framework, the simulations reproduced realistic mineralogical contrasts between quartz, feldspar, and mica grains, as well as the influence of intergranular contacts on fracture initiation and growth. The results highlight that fracture evolution follows a three-stage process, beginning with localized nucleation at high breakdown pressure, followed by stable extension guided by stress orientation and mineral interfaces, and culminating in accelerated branching and coalescence during the later stages of fluid injection.

A key finding is that weak mineral interfaces exert a fundamental control on fracture morphology. These interfaces not only reduce breakdown pressure but also promote deflection, branching, and network connectivity, thereby enhancing the potential for hydraulic connectivity within crystalline reservoirs. Stress field evolution further illustrates that principal stresses undergo significant redistribution during fracture development, with asymmetric relaxation caused by preferential propagation along weak planes. This behavior emphasizes the necessity of accounting for mineral-scale heterogeneity in order to achieve realistic predictions of fracture geometry and reservoir permeability.

The methodological contribution of this study lies in linking image-based microstructural characterization with discrete element simulations, enabling an explicit representation of heterogeneity that is often oversimplified in continuum approaches. Such integration provides valuable mechanistic insights into the role of grain boundaries and pre-existing discontinuities in guiding fracture development. Nevertheless, several limitations remain. The current simulations do not include thermal effects, which are critical in hot dry rock reservoirs, and the upscaling of microscale observations to field-scale models remains a major challenge.

Future work should focus on extending the modeling framework to fully coupled thermo-hydro-mechanical processes, improving the calibration of particle-scale parameters with experimental data, and integrating DEM results with reservoir-scale simulators to enable predictive modeling of long-term circulation performance. Field validation through microseismic monitoring and tracer tests will also be essential to confirm the reliability of microscale-based predictions. Ultimately, explicit consideration of mineral weak interfaces should be incorporated into stimulation design strategies, as their orientation and

distribution play a decisive role in controlling fracture complexity and reservoir productivity in granitic HDR systems.

## Acknowledgements

This work was also supported by the funding from the National Natural Science Foundation of China (No. U24B2034; No. U2139204), the Shanghai Rising Star Program (No. 24QA2709700), and the PetroChina Science and Technology Innovation Foundation (2021 DQ02-0501).

## Nomenclature

|               |                                                |
|---------------|------------------------------------------------|
| $a$           | = aperture of fracture or flow channel, m      |
| $E$           | = Young's modulus, Pa                          |
| $g$           | = gravitational acceleration, m/s <sup>2</sup> |
| $I$           | = moment of inertia, kg·m <sup>2</sup>         |
| $K_f$         | = fluid bulk modulus, Pa                       |
| $K_s$         | = shear stiffness of contact or bond, Pa       |
| $m$           | = particle mass, kg                            |
| $p$           | = fluid pressure, Pa                           |
| $R$           | = particle radius, m                           |
| $\sigma_{xx}$ | = horizontal stress in x-direction, Pa         |
| $\sigma_t$    | = tensile stress, Pa                           |
| $\theta$      | = friction angle, °                            |
| $u$           | = displacement, m                              |
| $V$           | = pore volume, m <sup>3</sup>                  |
| $\varphi$     | = porosity, dimensionless                      |
| $\rho$        | = density, kg/m <sup>3</sup>                   |
| $c$           | = cohesion, MPa                                |
| $f$           | = friction coefficient, dimensionless          |
| $h$           | = hydraulic aperture, m                        |
| $k$           | = permeability, m <sup>2</sup>                 |
| $K_n$         | = normal stiffness of contact or bond, Pa      |
| $L$           | = flow path length, m                          |
| $n$           | = unit normal vector, dimensionless            |
| $Q$           | = flow rate, m <sup>3</sup> /s                 |
| $\sigma$      | = stress, Pa                                   |
| $\sigma_{yy}$ | = horizontal stress in y-direction, Pa         |
| $\tau$        | = shear stress, Pa                             |
| $t$           | = time, s                                      |
| $v$           | = particle velocity, m/s                       |
| $\Delta P$    | = pressure difference, Pa                      |
| $\mu$         | = fluid viscosity, Pa·s                        |
| $\omega$      | = angular velocity, rad/s                      |

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